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Azmi Haider

Summary

PhD candidate in Generative AI and Computer Vision at the University of Haifa (Israel). My research focuses on generative modeling, probabilistic inference, and neural representations, with publications spanning 3D reconstruction, Neural Processes, Tokenized Neural Fields, and scientific machine learning. Seeking Research Scientist, Applied AI, Machine Learning, or Computer Vision roles worldwide.

Publications

A. Haider and D. Rosenbaum. **Direct Flow Neural Processes: Efficient Sampling via Flow Step Amortization.**
ICML 2026, Workshop on Structured Probabilistic Inference and Generative Modeling.

A. Haider and D. Rosenbaum. **Predicting 3D Structure by Latent Posterior Sampling.**
ICLR 2025, Workshop on Frontiers in Probabilistic Inference: Sampling Meets Learning.

A. Haider and D. Rosenbaum. **Tokenized Neural Fields: Structured Representations of Continuous Signals.**
NeurIPS 2025, Workshop on Structured Probabilistic Inference and Generative Modeling.

A. Haider and H. Hel-Or. **What Can We Learn from Depth Camera Sensor Noise?**
Sensors, 22(14):5448, 2022.

G. Alon, A. Haider, H. Hel-Or. **Judicial Errors: Fake Imaging and the Modern Law of Evidence**
21 UIC Rev. Intell. Prop. L. 82, 2022.

N. Privman-Horesh, A. Haider, and H. Hel-Or. **Forgery Detection in 3D-Sensor Images.**
CVPR 2018, Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops, 2018, pp. 1561-1569

D. Aibinder, A. Haider, and D. Rosenbaum. **Empirical Bayes Flow Matching for Continuous Cryo-EM Heterogeneity.**
Under submission (2026).

Professional Experience

2022 – 2023 | **Teaching Assistant** – University of Haifa, Israel
Taught Image Processing and Advanced Computer Vision courses to undergraduate students, leading tutorials, project mentoring, and office hours.

2021 – 2022 | **Data Science Intern (Face Recognition)** – Intel Corporation, Haifa, Israel
Developed image preprocessing, detection, and synthetic data generation pipelines for deep-learning-based face recognition systems.

2015 – 2021 | **Computer Vision Internships (3D Cameras & LiDAR)** – Intel Corporation, Haifa, Israel
Designed evaluation methodologies for 3D camera and LiDAR quality across 2D/3D sensing pipelines used in deployed computer vision systems.

Education

2022 – Present
Ph.D. in Computer Science – University of Haifa, Israel
Thesis: *Neural Representations for Probabilistic Inference and Generative Modeling*.
Advisor: Dr. Dan Rosenbaum.
Expected graduation: November 2026.

2017 – 2020
M.Sc. in Computer Science – University of Haifa, Israel
Thesis: *Forgery Detection in Depth Images*.
Advisor: Prof. Hagit Hel-Or.

2012 – 2017
B.Sc. in Electrical Engineering – Technion – Israel Institute of Technology
Focus: computer and software engineering.

Technical Skills

Programming: Python, PyTorch, OpenCV.

Generative Modeling: Diffusion models, flow matching, transformers, normalizing flows, variational autoencoders, autoregressive models.

Computer Vision: Implicit neural representations, 3D reconstruction, NeRF, Gaussian Splatting, probabilistic inference, representation learning.

Honors and Awards

2022 – 2025
Excellence Scholarship – University of Haifa

2022 – 2025
PhD Research Fellowship – Data Science Research Center, University of Haifa

2020
M.Sc. graduation with honors – University of Haifa

2011 – 2016
Full Scholarship (Outstanding Students Program) – Technion

Languages

Arabic — Native/Fluent
Hebrew — Fluent
English — Fluent
Spanish — Beginner